

NLP-Based Sentiment and Text Analysis of Gaming Platform Reviews

By

Lakmali Amarawardana

Objectives and approach

Gaming Platform Sentiment Analysis using NLP

- Analysis of user reviews and summaries

Understanding customer sentiment and experience

- Positive, Neutral, Negative ?
- Identify key themes using word frequency analysis.
- Compare emotional tone between reviews and summaries.

Converting insights into business actions

- Tailor-made customer segment treatment

Dataset Overview

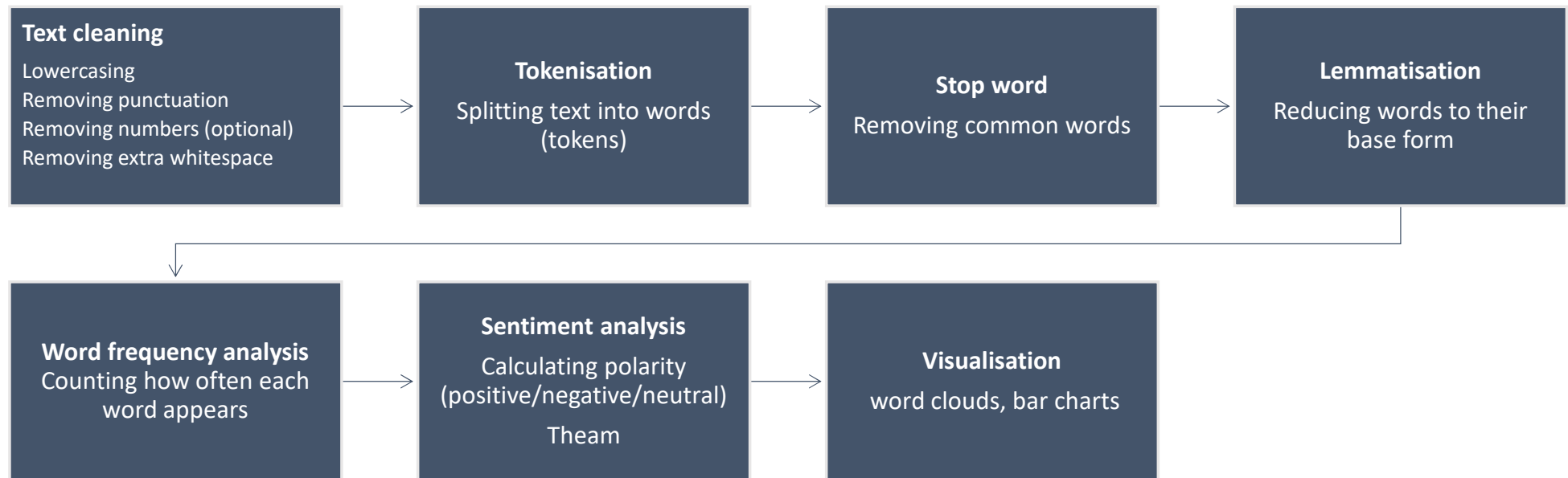


Reviews: detailed user feedback, unstructured



Summaries: compressed version of reviews

NLP Pipeline



Word Frequency Insights

Reviews

game (1941)

great (596)

fun (553), play (542)

Summaries

star (478)

five (381)

game (334)

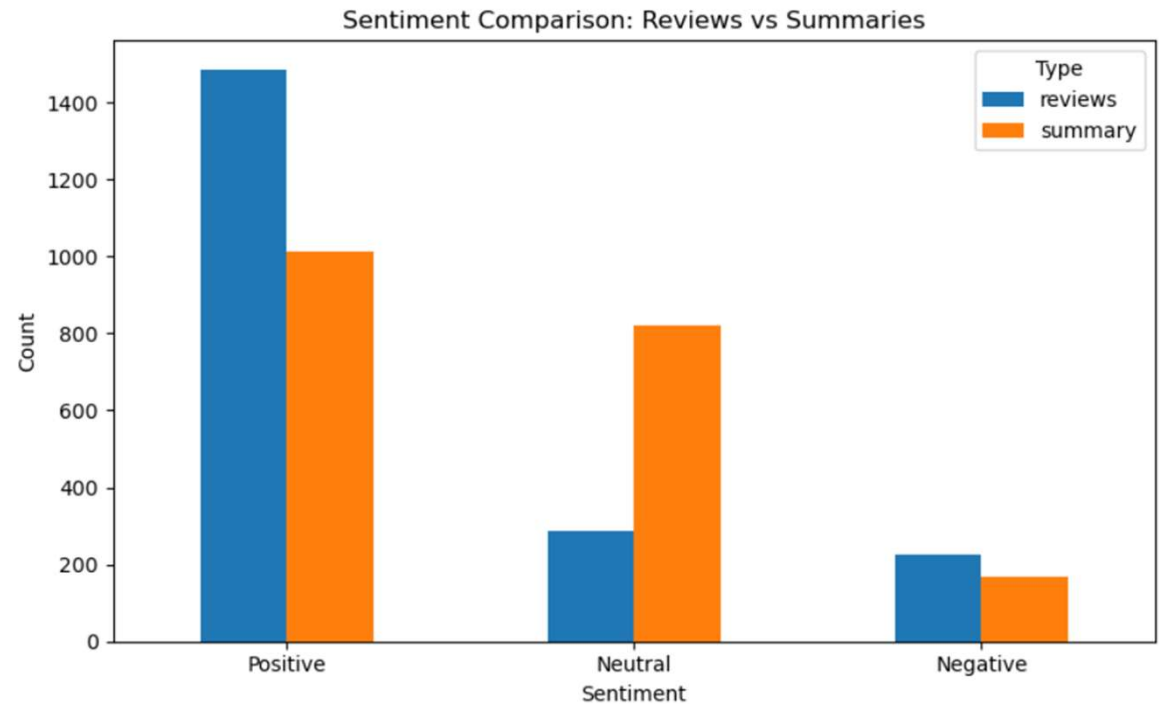
great (295), fun (218)

Insight: Focus on gameplay experience and rating-based feedback 😊

Sentiment Analysis Overview

Insights

- Overall sentiment is positive.
- Reviews exhibiting stronger emotional expression.
- Summaries contain a higher proportion of neutral sentiment.
- Negative sentiment remains relatively low in both cases



Customer Sentiment Overview

Key Findings

- Customers generally show **mildly positive sentiment**
- Mean sentiment $\approx$ **0.21 (Review & Summary)**
- Median is lower $\rightarrow$ many customers are **neutral to slightly positive**
- A few **extreme positive and negative outliers** exist

Key Insight

- Customer sentiment is **continuous, not strongly segmented**

Conclusion



Mildly positive
sentiment across
gaming platform



Reviews provide richer
emotional insight than
summaries



Summaries reduce
detail and emotional
intensity



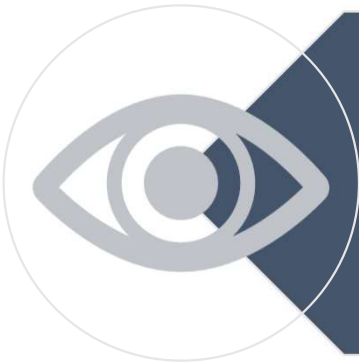
Relationship Between Reviews and Summaries

Evidence suggest that summaries may be derived from the original reviews; however, this requires further analysis to confirm the relationship between the two texts.

Analysis show,

- ✓ low lexical overlap
- ✓ compression in length
- ✗ not enough to prove derivation

Sentiment of K-Means Marginal Customers



Key Findings

- ~400 customers identified as **marginal (cluster boundary cases)**
- Their sentiment is **very similar to overall dataset**
- Slightly weaker sentiment, but not a distinct group



Key Insight

- Clusters are driven by **income & spending score**, not sentiment
- Sentiment alone does not explain cluster separation